

Big Data

Chapter 3

What are NoSQL databases?

Jorge Bernardino
jorge@isec.pt


Relational
NoSQL

Agenda

- A brief history of NoSQL databases
- What are NoSQL databases?
- Why NoSQL?
- How NoSQL Works?
- Features of NoSQL
- SQL vs NoSQL
- Key differences between SQL and NoSQL databases
- Key characteristics of NoSQL databases

Brief History of NoSQL Databases

The history of NoSQL databases is marked by the need for **scalable** and **flexible data storage solutions** to handle the challenges posed by the rapid growth of data in various industries.

Here is a brief timeline highlighting key developments and important dates in the history of NoSQL databases:

- **Late 1990s: Emergence of Non-relational Databases:**
  - Early experiments with non-relational databases, such as object-oriented databases, began in response to limitations of traditional relational databases in handling certain types of data and scaling challenges.
- **Early 2000s: Dynamo Paper (2007):**
  - Amazon released the "Dynamo" paper in 2007, describing a highly available and scalable key-value storage system. This paper laid the foundation for NoSQL databases by introducing concepts like partitioning and eventual consistency.

Giuseppe DeCandia et al. "Dynamo: Amazon's highly available key-value store." (2007): In Proceedings of twenty-first ACM SIGOPS symposium on Operating systems principles (SOSP '07) 205 220. https://doi.org/10.1145/1294261.1294281

Brief History of NoSQL Databases

- **2009: First NoSQL Meetup and the Term "NoSQL":**
  - The term "NoSQL" was coined during the first NoSQL Meetup in San Francisco in 2009. The event brought together engineers and developers interested in discussing and promoting non-relational databases.
- **2009: CouchDB and MongoDB Releases:**
  - CouchDB, a document-oriented database, and MongoDB, a document-oriented and scalable database, were released in 2009. These databases played a crucial role in popularizing the NoSQL movement.
- **2010: Apache Cassandra Becomes Open Source:**
  - Apache Cassandra, a highly scalable and distributed NoSQL database, was open-sourced by Facebook in 2010. Cassandra became known for its ability to handle large amounts of data across multiple nodes.

Brief History of NoSQL Databases

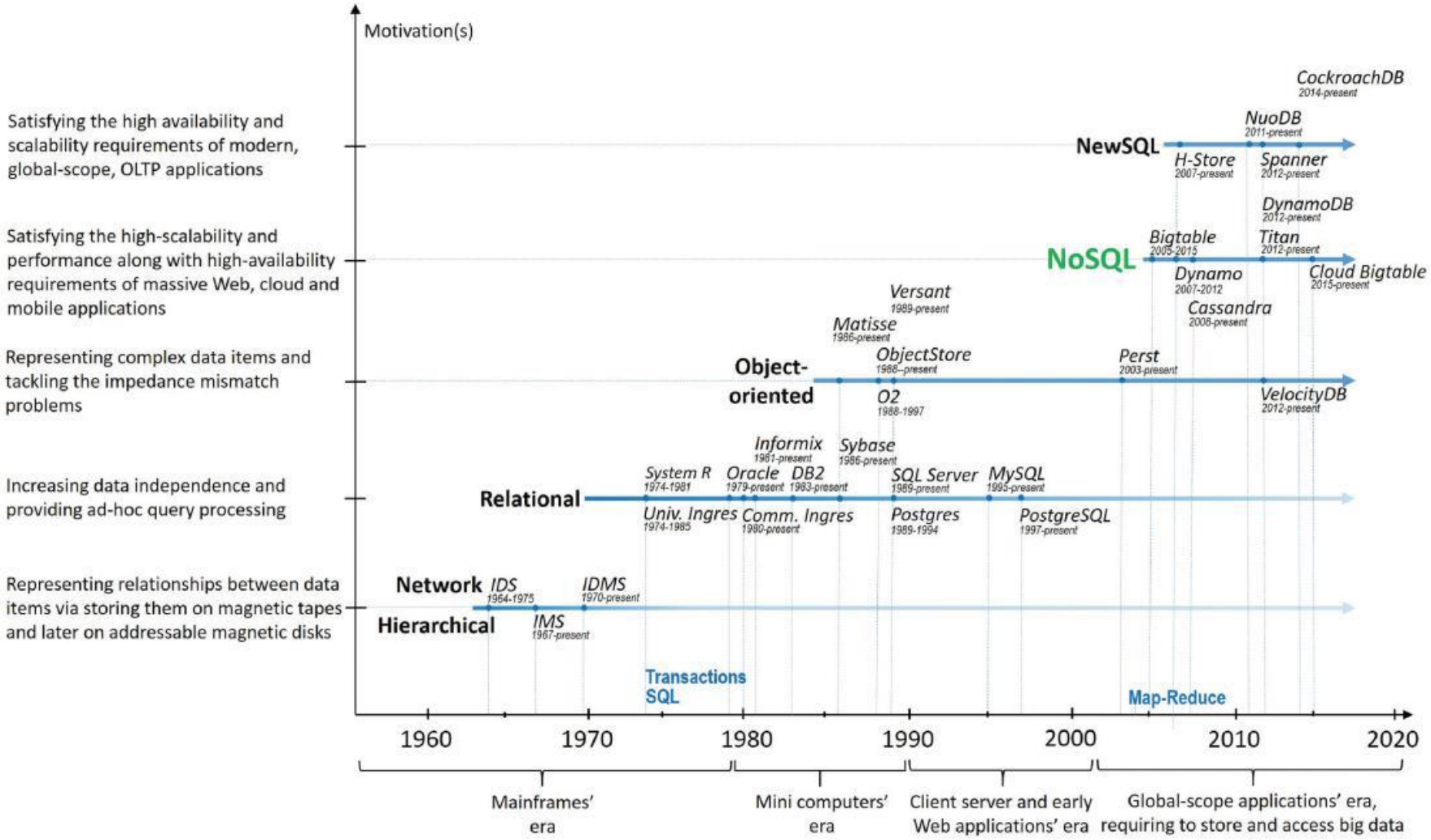
- **2012: MongoDB 2.2 Introduces Aggregation Framework:**
  - MongoDB 2.2 introduced the Aggregation Framework, providing a powerful and flexible way to perform data transformations and computations within the database.
- **2015: Rise of Graph Databases:**
  - The popularity of graph databases, such as Neo4j, increased in 2015. Neo4j, a graph database that uses the property graph model, gained recognition for its efficiency in managing and querying graph-structured data.
- **2016: NoSQL Database Survey by DB-Engines:**
  - NoSQL databases gained significant traction, and a 2016 survey by DB-Engines indicated a substantial increase in popularity for various NoSQL databases, with MongoDB, Cassandra, and Redis leading the rankings.

Brief History of NoSQL Databases

- **2017: Apache Kafka Gains Prominence:**
  - Apache Kafka, a distributed streaming platform, gained popularity for its role in handling real-time data streams and became widely used for building scalable and fault-tolerant event-driven architectures.
- **Present and Future: Continued Diversification and Adoption:**
  - The present landscape of NoSQL databases is characterized by a diverse range of solutions catering to different data models and use cases. NoSQL databases continue to evolve, with a focus on **performance**, **scalability**, and **ease of use**. The adoption of cloud-based NoSQL services has also become prevalent.

The history of NoSQL databases reflects the ongoing evolution of data management solutions in response to the changing needs of modern applications and the exponential growth of data in various industries.

The continuous development of major database technologies



IMS

The Hierarchical Database

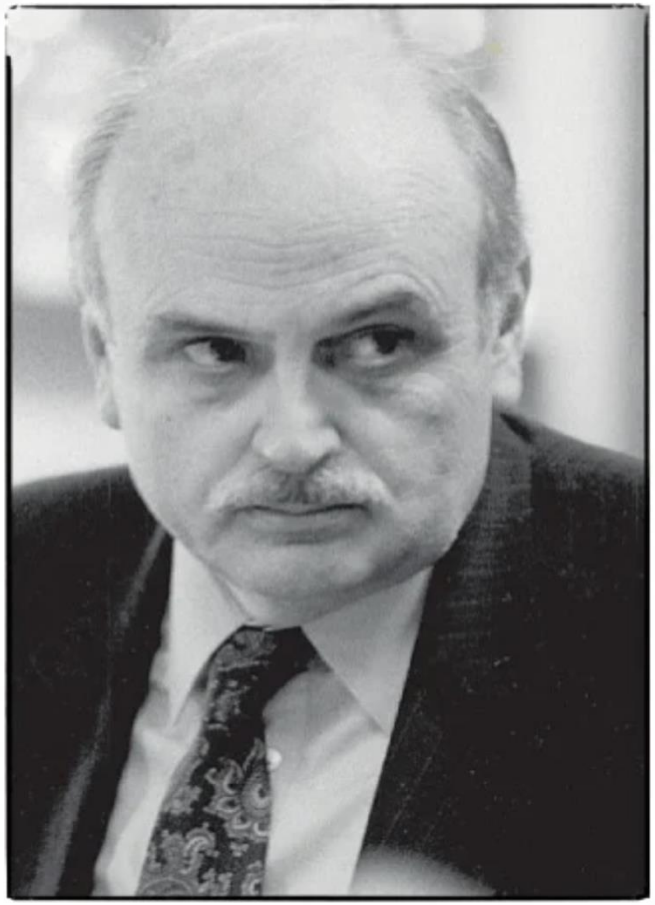
(1966)



Vern Watts

"A Relational Model for Large Shared Databanks"

(1970)



Ted Codd

""SEQUEL: A Structured English Query Language"

Don Chamberlin & Ray Boyce

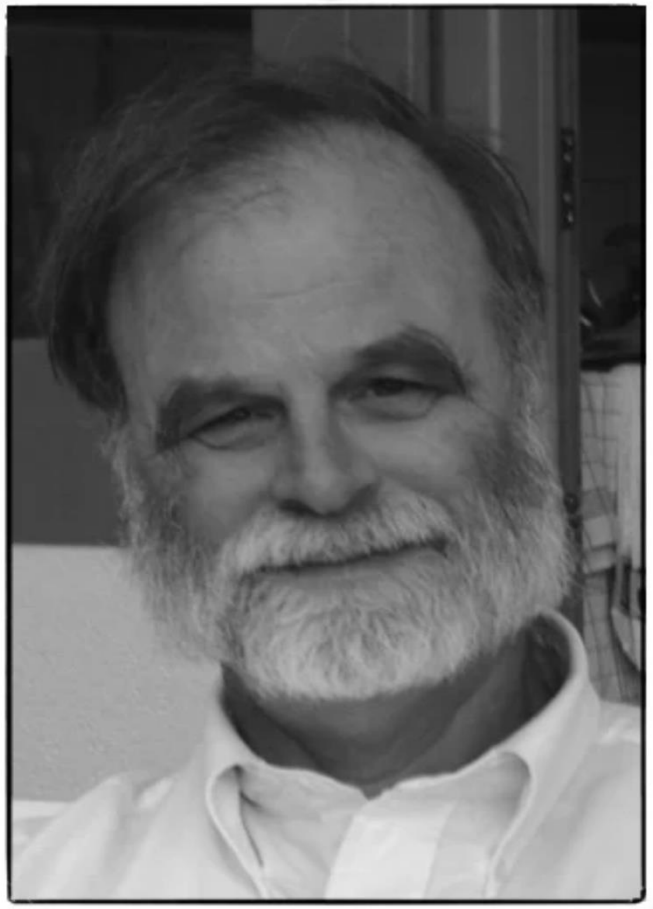
(1974)



Don Chamberlin

ACID

(late 1970's)



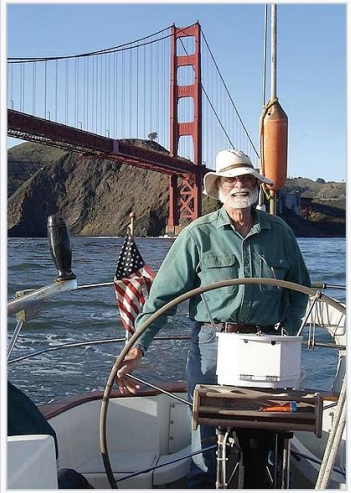
Jim Gray

Jim Gray

- https://en.wikipedia.org/wiki/Jim_Gray_(computer_scientist)

Disappearance [edit]

Gray, an experienced sailor, owned a 40 foot (12 m) sailboat. On January 28, 2007, he failed to return from a short solo trip to scatter his mother's ashes at the Farallon Islands near San Francisco.[20] The weather was clear, and no distress call was received, nor was any signal detected from the boat's automatic Emergency Position-Indicating Radio Beacon.



Jim Gray on his yacht *Tenacious* in 2006

A four-day Coast Guard search using planes, helicopters, and boats found nothing.[21][22][23][24] On February 1, 2007, the DigitalGlobe satellite scanned the area[25] and the thousands of images were posted to Amazon Mechanical Turk. Students, colleagues, and friends of Gray, and computer scientists around the world formed a "Jim Gray Group" to study these images for clues. On February 16 this search was suspended,[26] and an underwater search using sophisticated equipment ended May 31.[9][27][28][29][30]

Marine search expert Bob Bilger explains that the type of boat used by Gray, a C&C 40 is vulnerable to hull damage, and can sink as quickly as in 30 seconds, also taking any equipment with it, not leaving any loose debris.[31]

The University of California, Berkeley and Gray's family hosted a tribute on May 31, 2008.[32] Five years after the disappearance, Carnes petitioned a court to have her husband declared dead and on January 28, 2012, Gray was declared legally dead.[33][34][35]

In 2012, Carnes co-authored a paper on coping with ambiguous loss.[36] In 2014, in conjunction with the disappearance of Malaysia Airlines Flight 370, CNN interviewed Carnes.[35] At the time, she lived with her elderly mother suffering from dementia in Wisconsin.

isec Engenharia

"NoSQL" coined

(1998)



Carlo Strozzi



"NoSQL" Reintroduced

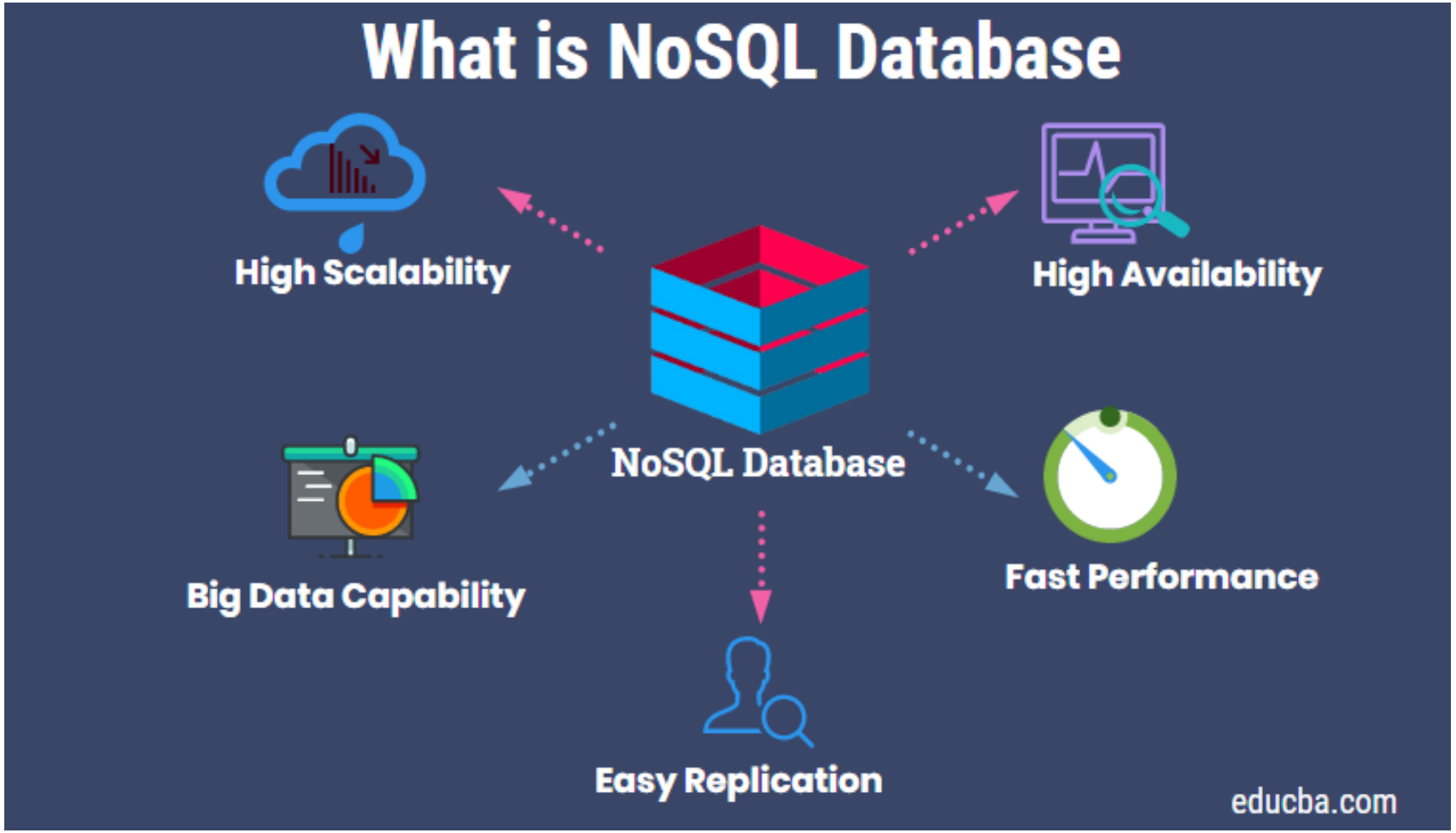
(2008)

NoSQL

- The term NoSQL actually comes from a Twitter hashtag that was suggested by Eric Evans for the purpose of a meeting in 2008 to discuss big data and linearly scalable distributed systems.
- Johan Oskarsson was organizing the first meetup and asked the question "What's a good name?" on IRC; it was one of 3 or 4 suggestions that I spouted off in the span of like 45 seconds, without thinking.
- Since then, the term has almost taken on a life of its own.


HBASE
CouchDB
relax
riak
Cassandra
Neo4j
the graph database
mongoDB
Redis
Tokyo
Cabinet 8192PiB
InfiniteGraph

What is a NoSQL database?



NoSQL Definition

NoSQL is an approach to database design that provides **flexible schemas** for storing and retrieving data beyond the traditional table structures found in relational databases.

Instead of the typical tabular structure of a relational database, NoSQL databases house data within one data structure. Because this non-relational database design does not require a schema, it provides **rapid scalability** to manage large and typically **unstructured data** sets.

What is NoSQL?

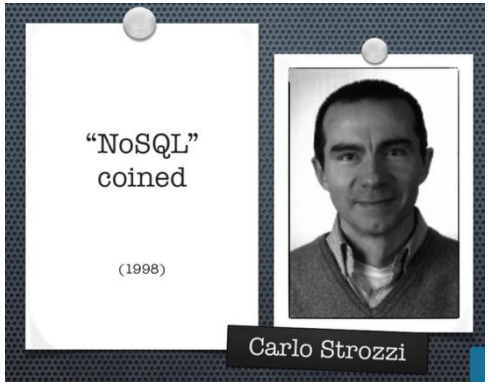
- NoSQL (Not SQL or Not Only SQL) is a generic term used for databases that do not depend on a relational model.
- The data does not need to have a strict schema nor the usual SQL table structure.
- Most commonly, the data is aggregated as **key-value pairs, JSON documents, graphs,** or **wide-column tables.**

What is NoSQL?

- NoSQL Database is a non-relational DBMS
  - Does not require a fixed schema.
  - It avoids joins and is easy to scale.
- NoSQL is used for Big Data and real-time web apps.
- For example, companies like Twitter(X), Facebook and Google collect terabytes of user data every single day.

NoSQL

- **NoSQL** stands for "Not Only SQL" or "Not SQL."
- Though a better term would be "NoREL".
- Carl Strozzi introduced the NoSQL concept in 1998.



NoSQL

- NoSQL databases are designed to handle large volumes of unstructured, semi-structured, or structured data, providing flexible and scalable storage solutions.

- Unlike relational databases, NoSQL databases do not rely on a fixed schema, allowing for more dynamic and agile data management.



Why NoSQL?

- The need to capture and process big data is the main driving force behind NoSQL database popularity.
- The stored data must be accessible at all times, from any location, and on any device.
- One way to meet the growing demand is to scale up and buy a bigger server. However, it is more efficient to scale out and use a cluster of on-demand servers.

Why NoSQL?

- The relational database model is not well-suited for a distributed system spanning multiple machines.

- NoSQL databases provide a viable solution by focusing on performance and availability while also sacrificing some of the consistency usually identified with relational databases.

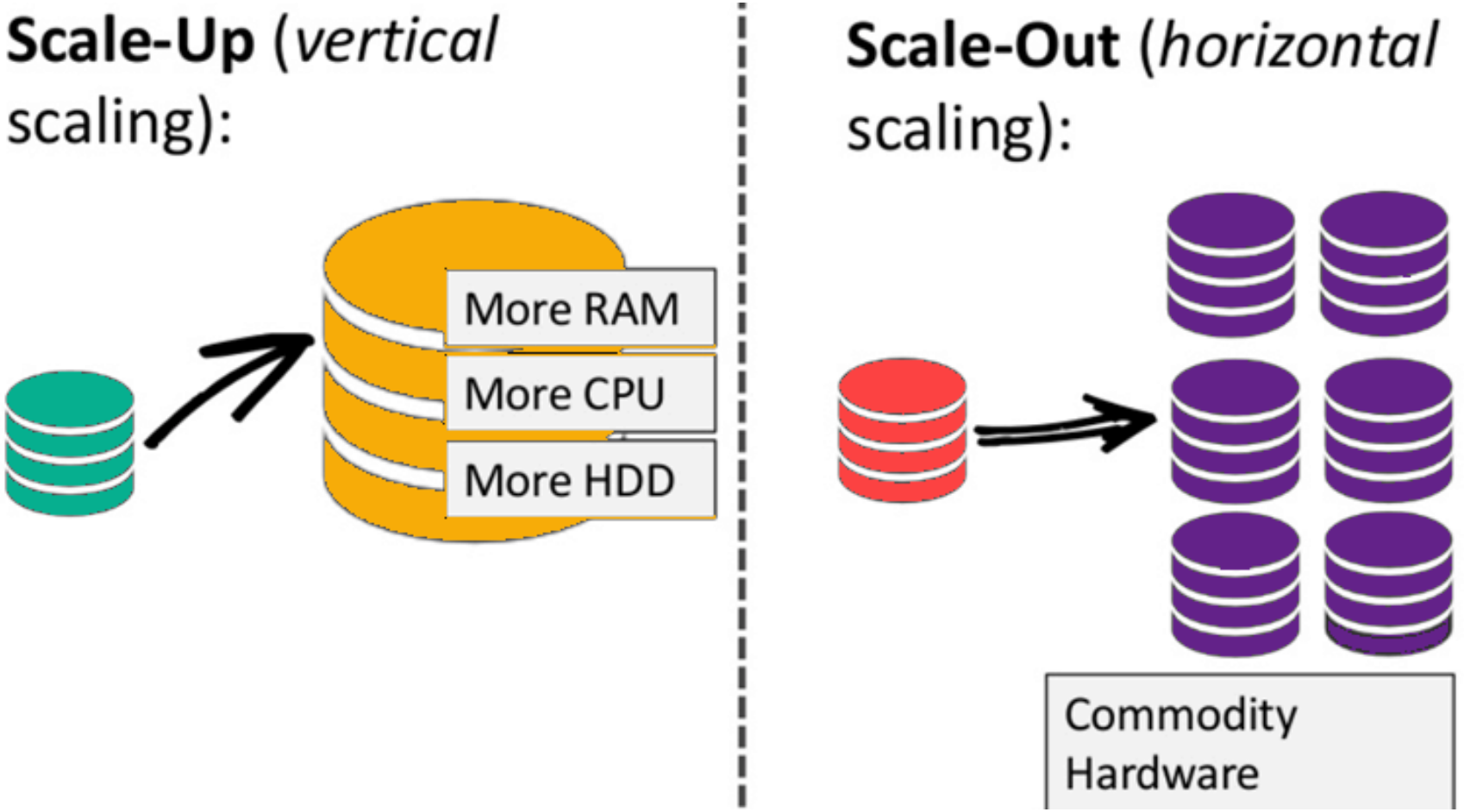
Why NoSQL?

- The concept of NoSQL databases became popular with Internet giants like Google, Facebook, Amazon, etc. who deal with huge volumes of data. The system response time becomes slow when we use RDBMS for massive volumes of data.

- To resolve this problem, we could "scale up" our systems by upgrading our existing hardware.

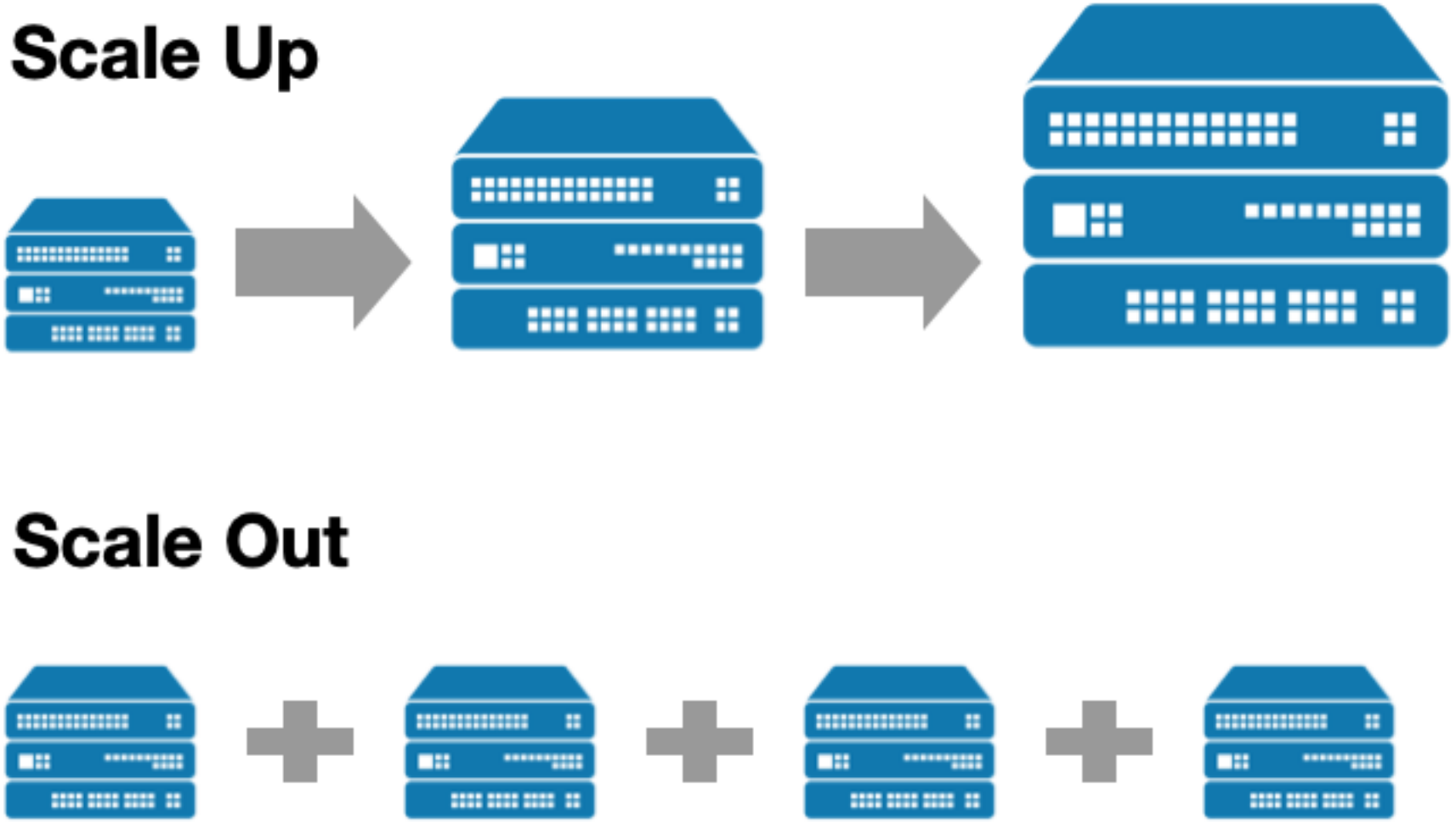
- The alternative for this issue is to distribute database load on multiple hosts whenever the load increases. This method is known as "scaling out".

Scale-Up (**vertical**) vs Scale-Out (**horizontal**)



NoSQL database is non-relational, so it scales out better than relational databases

Scale-Up (**vertical**) vs Scale-Out (**horizontal**)



Features of NoSQL

- Non-relational
- Schema-free
- Simple API
- Distributed

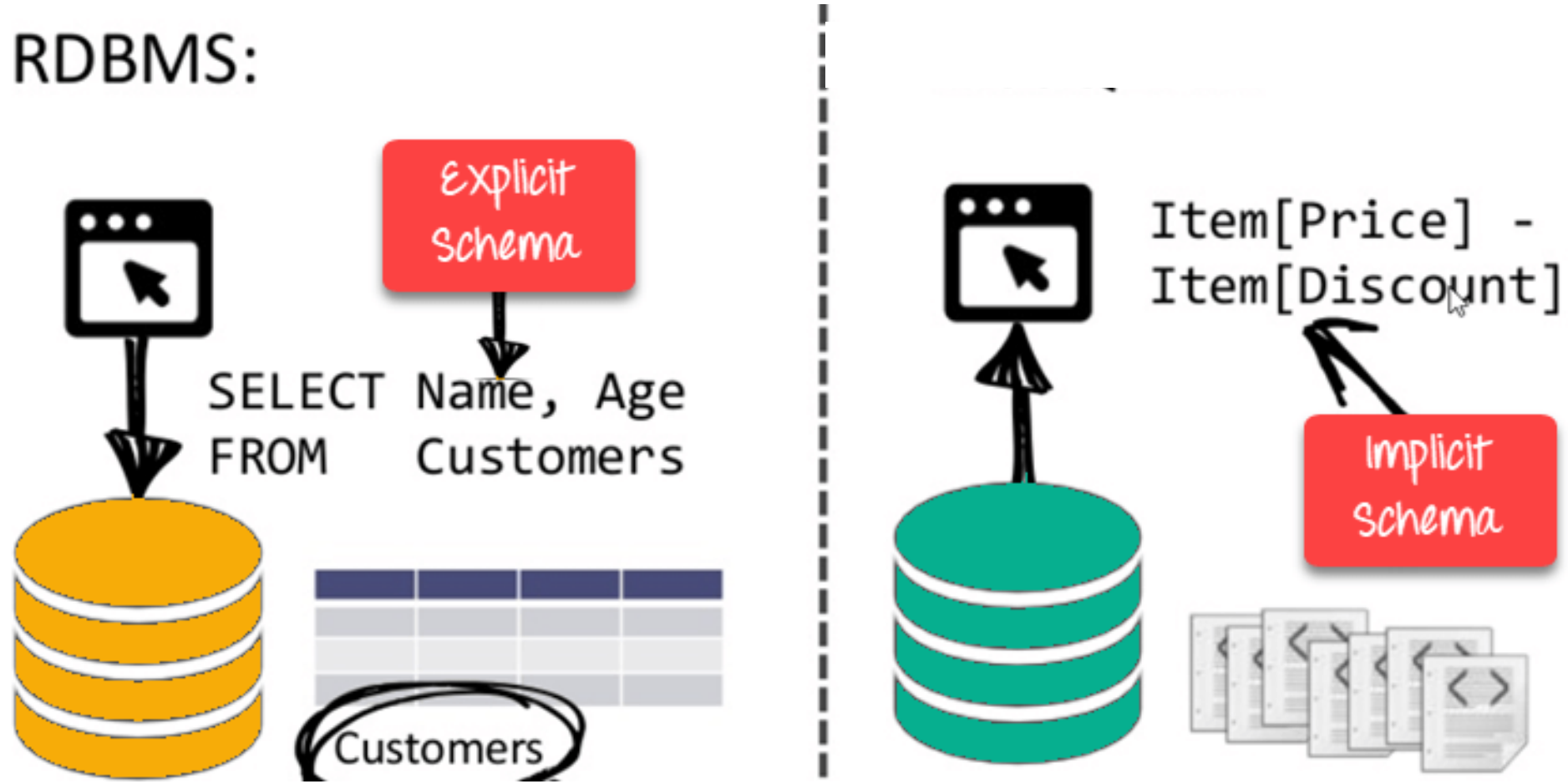
Features of NoSQL

- **Non-relational**
  - NoSQL databases never follow the relational model
  - Never provide tables with flat fixed-column records
  - Work with self-contained aggregates or BLOBs
  - Doesn't require object-relational mapping and data normalization
  - No complex features like query languages, query planners, referential integrity joins, ACID.

Features of NoSQL

- **Schema-free**
  - NoSQL databases are either schema-free or have relaxed schemas
  - Do not require any sort of definition of the schema of the data
  - Offers heterogeneous structures of data in the same domain.

NoSQL is Schema-Free



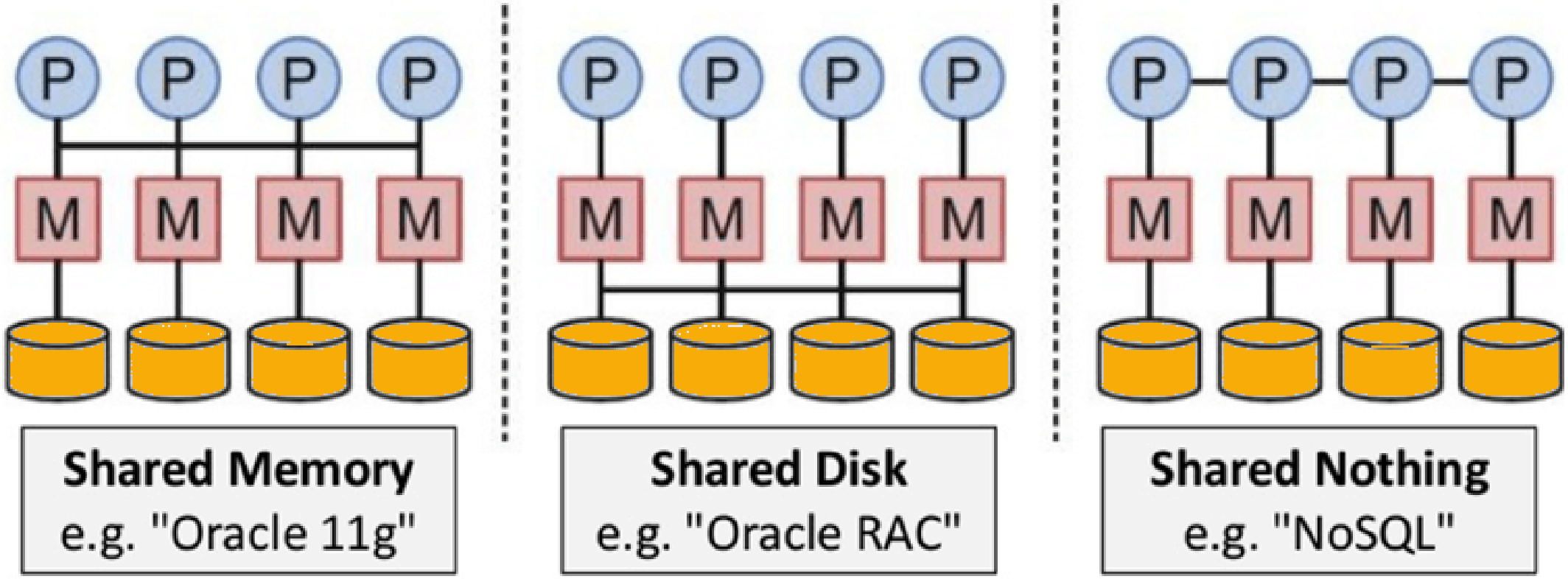
Features of NoSQL

- **Simple API**
  - Offers easy to use interfaces for storage and querying data provided
  - APIs allow low-level data manipulation and selection methods
  - Text-based protocols mostly used with HTTP REST with JSON
  - Mostly used no standard based NoSQL query language
  - Web-enabled databases running as internet-facing services.

Features of NoSQL

- **Distributed**
  - Multiple NoSQL databases can be executed in a distributed fashion
  - Offers auto-scaling and fail-over capabilities
  - Often ACID concept can be sacrificed for scalability and throughput
  - Mostly no synchronous replication between distributed nodes Asynchronous Multi-Master Replication, peer-to-peer, HDFS Replication
  - Only providing eventual consistency
  - Shared Nothing Architecture. This enables less coordination and higher distribution.

Shared


P
M
Shared Memory
e.g. "Oracle 11g"
Shared Disk
e.g. "Oracle RAC"
Shared Nothing
e.g. "NoSQL"

How NoSQL Works?

- The basic idea behind NoSQL is to optimize the database performance for horizontal scaling, large data volumes, and low latency by sacrificing some data consistency restrictions present in RDBMS.
- Instead of rigid data models such as tables, columns, or rows, NoSQL databases offer flexible models.
- In use cases that do not require relational consistency, these models help NoSQLs perform better than relational databases.

Features of NoSQL Databases

- NoSQL databases are structurally diverse and offer various data storage models.
- There are, however, several common attributes that distinguish NoSQL from relational databases.



Features of NoSQL Databases

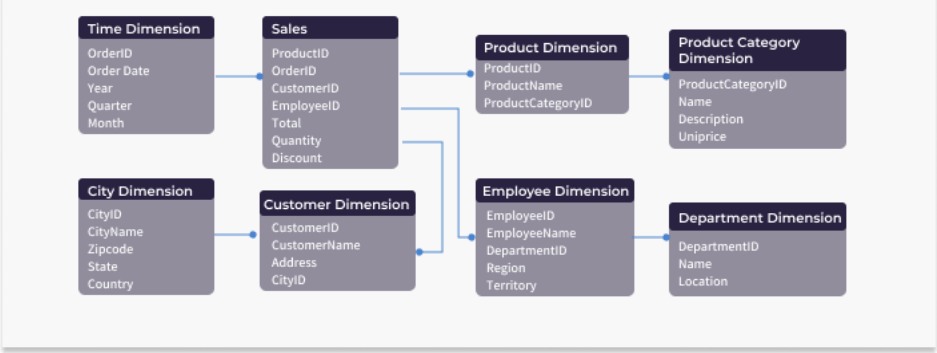
Schema on Read

- A NoSQL database allows to store data before applying a **structure** or **schema**.
- The schema is applied by the application code only when it accesses data. This process is often referred to as ***schema on read*.**
- By not structuring data in advance, NoSQL databases can write and read immense volumes of data significantly faster than a relational database can.

Features of NoSQL Databases

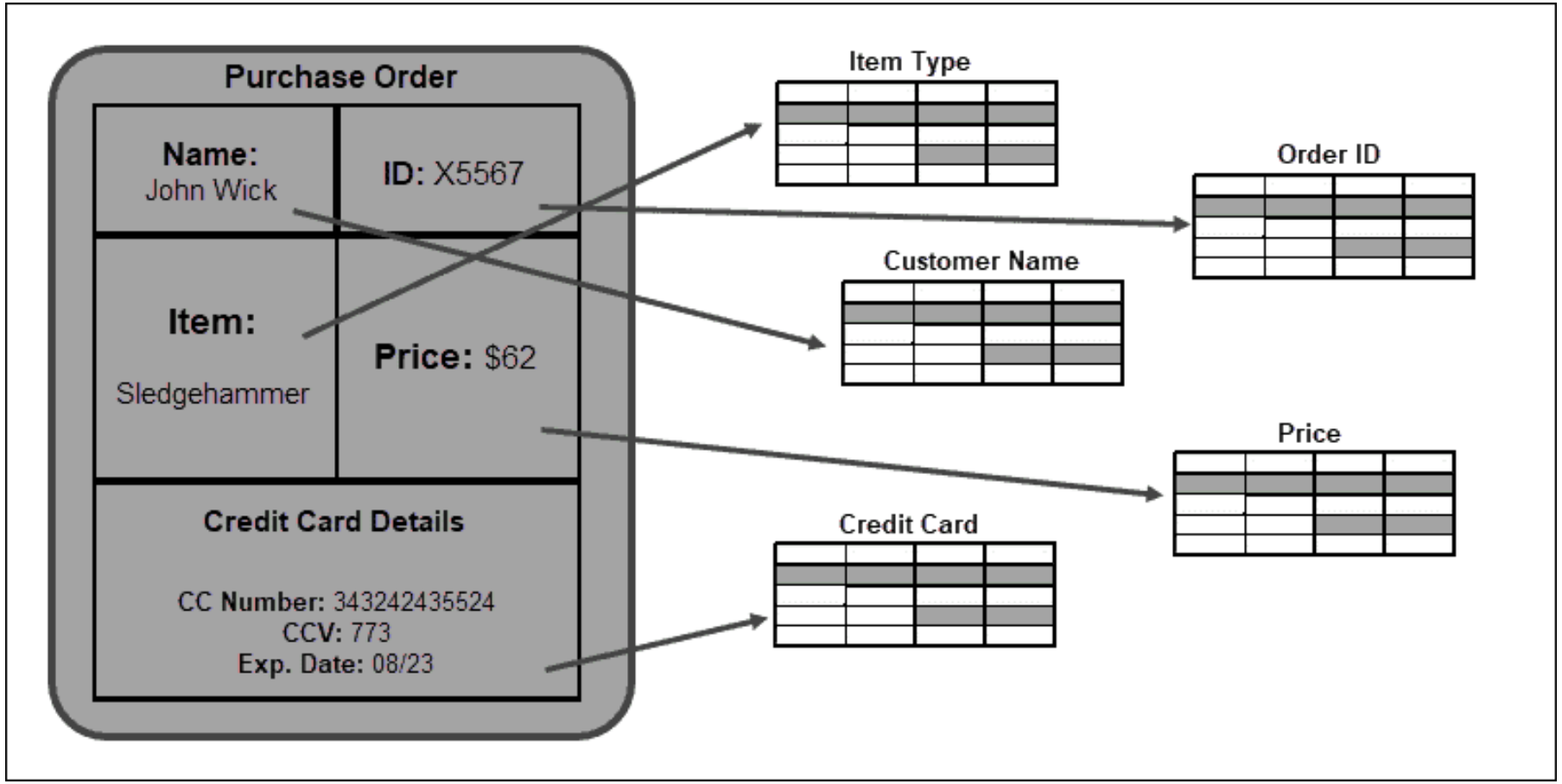
NoSQL vs Relational Databases

- In contrast, an SQL relational model, structures incoming data before it is written to a database.
- Predefined schema design is used to classify all the possible data types in advance.
- The schema is applied as data is structured and stored within tables, columns, and rows.

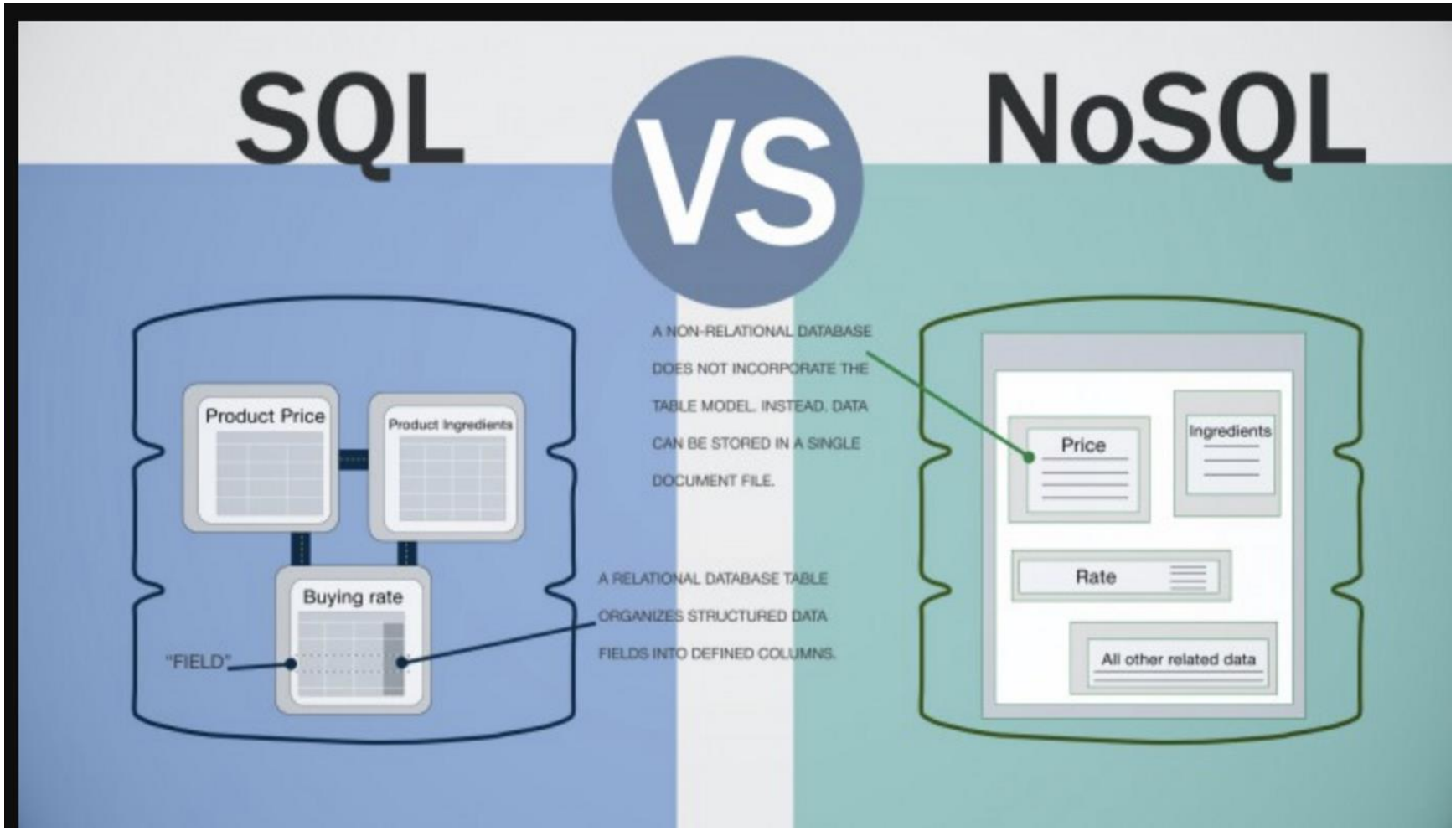


Features of NoSQL Databases

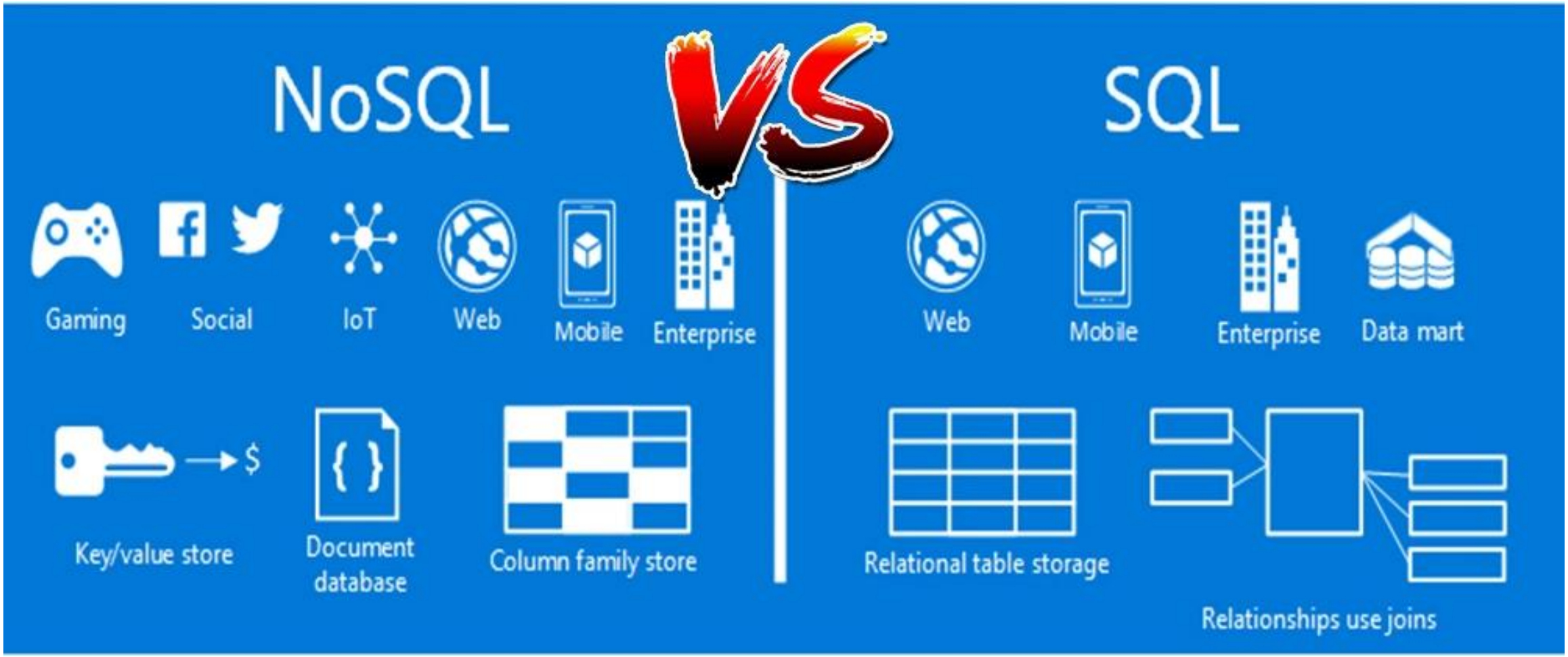
NoSQL vs Relational Databases



SQL vs NoSQL


SQL
VS
NoSQL
Product Price
Product Ingredients
Buying rate
"FIELD"
A RELATIONAL DATABASE TABLE
ORGANIZES STRUCTURED DATA
FIELDS INTO DEFINED COLUMNS.
A NON-RELATIONAL DATABASE
DOES NOT INCORPORATE THE
TABLE MODEL. INSTEAD, DATA
CAN BE STORED IN A SINGLE
DOCUMENT FILE.
Price
Ingredients
Rate
All other related data

SQL vs NoSQL


NoSQL
VS
SQL
Gaming
Social
IoT
Web
Mobile
Enterprise
Web
Mobile
Enterprise
Data mart
Key/value store
Document database
Column family store
Relational table storage
Relationships use joins

SQL vs NoSQL

Comparing	SQL	NoSQL
Query Language	Structured query language (SQL)	No declarative query language
Database type	Table	Key-value, document, wide-column, and graph
Schema	Predefined	Dynamic
Data model	Relational	Non-relational
Popular database management systems	MySQL, PostgreSQL, Oracle, and MS-SQL	MongoDB, Apache HBase, Amazon DynamoDB, Redis, Couchbase, Cassandra, and Elasticsearch
Ability to scale	Vertical	Horizontal
Hardware	Specialized DB hardware (Oracle Exadata, etc.)	Commodity hardware
ACID vs BASE	ACID	BASE
Open-source	A mix of open-source like Postgres & MySQL, and commercial like Oracle Database.	Open-source
Advantages	Cross-platform support, secure and free	Easy to use, high performance, and flexible tool
Disadvantages	Complex to maintain and inefficient if processing big data, complex relational database systems are difficult to export into other systems, not good for handling various data types	Data is less structured, NoSQL databases are not as reliable (no ACID support), NoSQL databases are newer and may offer less features than their SQL counterparts
Use cases	ACID support, complex queries, no changes or growth	Real-time data, volumes of data with no structure, agile business, cloud computing

5 Key differences between SQL and NoSQL databases

1. **Data Model**
2. **Schema**
3. **Scalability**
4. **Consistency and ACID Properties**
5. **Query Language.**

5 Key differences between SQL and NoSQL databases

isec Engenharia

1. Data Model

- **SQL**: Relational databases use a structured, tabular data model with predefined schemas. Data is organized into tables with rows and columns, and relationships between tables are established using foreign keys.
- **NoSQL**: NoSQL databases support various data models, including document-oriented (e.g., MongoDB), key-value pairs (e.g., Redis), column-family (e.g., Cassandra), and graph-oriented (e.g., Neo4j). NoSQL databases offer more flexibility in terms of data structure and schema.

5 Key differences between SQL and NoSQL databases

2. Schema

- **SQL**: Relational databases have a fixed schema, meaning that the structure of the database, including tables and their relationships, must be defined before inserting data.
- **NoSQL**: NoSQL databases typically have a dynamic or schema-less approach, allowing for on-the-fly changes to the data structure without requiring a predefined schema. This flexibility is beneficial in handling diverse and evolving data.

5 Key differences between SQL and NoSQL databases

isec Engenharia

3. Scalability

- **SQL**: Traditional relational databases often scale vertically, meaning that to handle increased load, more powerful hardware is added to a single server. This approach has limitations.
- **NoSQL**: NoSQL databases are designed to scale horizontally by adding more servers to a distributed network. This allows for better handling of large volumes of data and increased performance.

5 Key differences between SQL and NoSQL databases

4. Consistency and ACID Properties

- **SQL**: Relational databases adhere to the ACID properties (Atomicity, Consistency, Isolation, Durability), providing strong consistency and ensuring that transactions are reliably processed.
- **NoSQL**: NoSQL databases may sacrifice strict consistency for performance and availability, often following the CAP theorem (Consistency, Availability, Partition tolerance). Different NoSQL databases may prioritize different aspects of the CAP theorem.

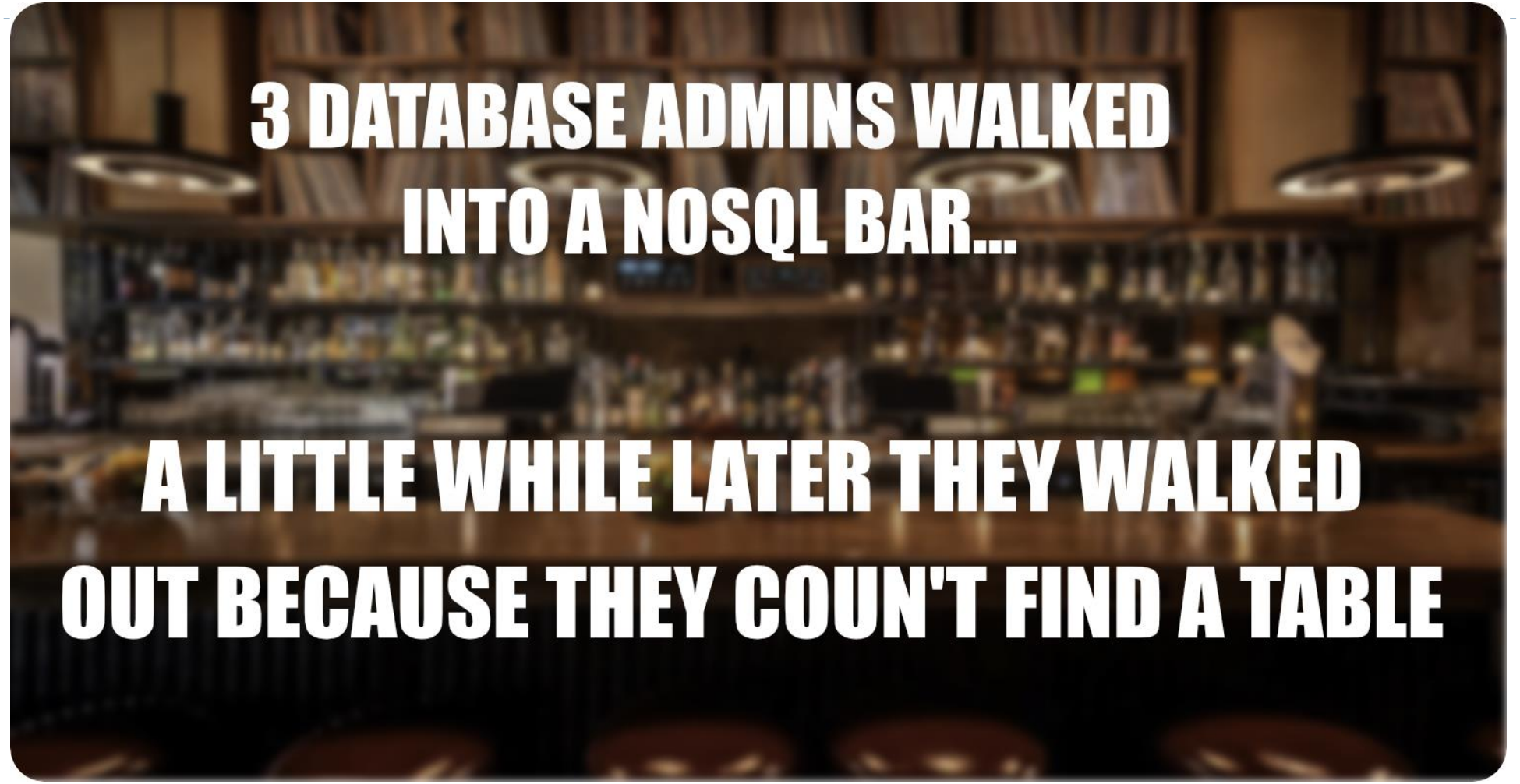
5 Key differences between SQL and NoSQL databases

5. Query Language

- **SQL**: Uses a standardized query language (SQL) for defining and manipulating relational databases. SQL allows for complex queries, joins, and transactions.
- **NoSQL**: Each type of NoSQL database may have its query language or API. The query languages are often designed to be more specific to the data model used by the database (e.g., MongoDB's query language for document databases).

SQL vs NoSQL

Feature	SQL and NoSQL database difference	
	SQL	NoSQL
Data model	Structured	Unstructured or semi-structured
Scalability	Vertical scaling	Horizontal scaling
Performance	Complex queries can be slow	Fast read and write performance
ACID compliance	Yes	No or partial
Schema	Rigid	Flexible
Language	SQL (Structured Query Language)	JSON (JavaScript Object Notation), XML, YAML, or binary schema


3 DATABASE ADMINS WALKED
INTO A NOSQL BAR...
A LITTLE WHILE LATER THEY WALKED
OUT BECAUSE THEY COUN'T FIND A TABLE

4 Key characteristics of NoSQL Databases

1. Schema Flexibility

- NoSQL databases allow for dynamic and flexible data models, accommodating various data formats without a predefined schema. This flexibility is particularly beneficial when dealing with diverse and evolving data types.

2. Scalability

- NoSQL databases are designed to scale horizontally, distributing data across multiple servers or nodes. This enables efficient handling of large volumes of data and supports high read and write throughput.

4 Key characteristics of NoSQL Databases

3. Performance

- NoSQL databases often prioritize performance and low-latency access to data. They are well-suited for applications requiring quick and agile data retrieval, especially in scenarios with high-velocity data.

4. Distributed Architecture

- Many NoSQL databases are built with distributed architectures, allowing them to operate efficiently across clusters of machines. This supports high availability and fault tolerance.

Summary

- NoSQL is a non-relational DMS, that does not require a fixed schema, avoids joins, and is easy to scale
- The concept of NoSQL databases became popular with Internet giants like Google, Facebook, Amazon, etc. who deal with huge volumes of data
- In the year 1998- Carlo Strozzi use the term NoSQL for his lightweight, open-source relational database
- NoSQL databases never follow the relational model it is either schema-free or has relaxed schemas
- 5 Key differences between SQL and NoSQL databases
- 4 Key characteristics of NoSQL Databases.


Any Questions ?